Sustainability Aware Asset Management

Emerging Markets, Scopes 1+2

Group Project

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Abstract

The main goal of this project is to study whether it is possible to introduce carbon-aware constraints without reducing the financial portfolio performance. The analysis uses a filtered universe of emerging market equities from 2014 to 2025, with both financial and environmental data. The first section of this project consists on evaluating two different types of portfolios that will be used as a reference for the rest of the report. The study computes a long only value-weighted benchmark with a minimum-variance portfolio; portfolio allocation uses a rolling 10-year window of monthly returns. These portfolios are then evaluated through their volatility, annualized return, cumulative return and Sharpe Ratio. The results show that the minimum-variance portfolio achieves lower volatility but with higher risk-adjusted returns, while the value-weighted benchmark delivers the best absolute performance. Sustainability constraints are then introduced in the second part of this project. New allocation frameworks include a minimum-variance portfolio with a 50% ecological impact reduction, a tracking error minimization portfolio also with the 50% emissions reduction and a more strict strategy with a net-zero allocation that imposes a 10% annual reduction of the carbon footprint from the first observation. Portfolios' carbon performances are then assessed using carbon intensity measures and footprint. Overall, the project finds that the introduction of sustainability constraints preserve substantial financial performance while reducing carbon emission of the portfolio. These findings support the claim that de-carbonization and portfolio efficiency can coexist in the emerging market equities world.

Keywords: Sustainability, portfolio optimization, minimum-variance portfolio, carbon footprint reduction, net-zero investing, emerging markets

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1 Introduction

Climate change is becoming a major factor affecting global financial markets, forcing investors to account for environmental impacts in portfolio construction. In current markets caring about carbon emissions is no longer just an optional feature, but many institutional investors now face some ESG constraints imposed by regulators and stakeholders. Driven by new frameworks like the EU Sustainable Finance Disclosure Regulation (SFDR) and the Task Force on Climate-related Financial Disclosures (TCFD), controlling carbon emissions is no longer a "bonus" feature but rather a mandatory aspect of risk management.

The main objective of this project is to analyze the effect of carbon emissions constraints on portfolio optimization, evaluating the trade off between performance and carbon emissions reductions, specifically within the universe of Emerging Markets (EM) equities. It is particularly interesting analyzing emerging markets as they are often characterized by higher carbon intensities and a heavy dependence on fossil fuels. Using data on Scope 1 and Scope 2 greenhouse gas emissions and monthly stock returns, the project evaluates whether it is possible to highly, if not completely, decarbonize an EM portfolio without sacrificing financial performance.

The analysis relies on the creation of an investment universe by processing data of a broad sample of EM equities spanning from 2000 to 2025. To avoid look-ahead bias and survivorship bias, strict data cleaning processes are implemented. This includes dynamic 10-year rolling window for asset selection, strict delisting logic (recognizing -100% returns upon firm exit), and liquidity filters to remove stale prices. The portfolio optimization problems are then simulated strictly out-of-sample over a 144-month period from January 2014 to December 2025.

The report is structured into three main stages:

- **Standard Portfolio Allocation:** Two basic strategies are established to be used as benchmarks: a long-only minimum-variance portfolio ($P_{oos}^{(mv)}$) and a value-weighted benchmark ($P^{(vw)}$).
- **Portfolio Allocation with Carbon Emission Reduction:** A strict 50% carbon footprint reduction constraint is applied. In particular, two portfolios are evaluated: an active carbon-constrained minimum-variance portfolio, and a passive tracking error minimization portfolio that closely tracks the value-weighted benchmark while halving its ecological impact.
- **Portfolio Allocation with a Net Zero Objective:** Finally, a forward-looking Net Zero trajectory strategy is constructed. This portfolio imposes a compounded 10% annual reduction in the portfolio's carbon emissions starting from a 2013 baseline.

Overall, the findings of this project demonstrate that strong sustainability constraints can be successfully applied to emerging markets. In particular, by moving capital away from high-emission sectors, the optimized portfolios obtain high decarbonization while maintaining strong performance against the benchmarks. These results strongly confirm the idea that carbon-aware investing and portfolio efficiency can coexist, providing empirical evidence in emerging markets for the ESG-efficient frontier model proposed by Pedersen et al. (2021).

2 Data Cleaning and Processing

The dataset is composed by financial and environmental data for Emerging Markets firms from 1999 to 2025. The raw data are downloaded from the Datastream platform and require several cleaning steps before doing any portfolio construction. The raw returns matrix, once loaded and transposed contains 314 monthly observations and 668 firms, before any dynamic filtering is applied. The cleaning process is organized in five steps: removing invalid Datastream identifiers, treating missing price observations, filtering out low and stale prices, preparing the carbon emissions data and lastly, building the dynamic investment set. Each step is designed to produce a dataset that is free of look-ahead bias, consistent and ready for the portfolio optimization that follows.

2.1 Datastream Error Rows and Standardisation

Some rows in the raw Datastream files are marked with `"$$ER"`, indicating that no valid data could be found for that security, for example because of missing data for that market. These rows contain useless information and are removed from all datasets before the analysis. After removing the error rows, each file goes through a standardisation step. ISIN codes are converted into a consistent format to ensure uniformity across the prices, market capitalisation, revenue and carbon data files. Non-financial columns added by Datastream during the export are dropped. Each file is then transposed so that dates become the Row and firm identifiers become the Columns index, all remaining values are converted to numeric format. This ensures that all matrix operations and data selections follow the same time structure.

2.2 Missing Values

Missing values in the price index series can happen for three reasons. The first case applies for the firms that are not yet listed or they are not yet reported at the beginning of the sample. In this case, no adjustment is needed: the missing values are left as `NaN` and the firm is just excluded from the investment set until valid data becomes available. Filling these observations would mean assigning a price history to a firm before it even existed, by doing so we introduce look-ahead bias into the covariance matrix. The second case applies to the gaps that occur in the middle of the sample period, usually they reflect a temporary reporting problem, data errors or short trading suspensions. A forward filling procedure is used to cover these missing values, the most recent valid price is carried forward until a valid price appear. This maintains the continuity of the return series without using any information that would not have been available at the time of the gap. The third case applies to the missing observations at the end of the firm sample, which they usually correspond to de-listings, default or other types of market exits. To avoid survivorship bias and to prevent the covariance matrix from underestimating the actual risk of the investment set, we took the following steps. The first step is to identify the last valid price date before any forward-filling. A return of -100% is assigned to the month after the final valid price to reflect the economic loss associated with a firm exit. All later months are set to `NaN` to avoid artificial zero-variance observations to subsequent estimation windows. These three treatments ensure that the return series reflects a realistic picture of how firms enter and exit the market, rather than an artificially clean history which would make the investment set appear more stable than it actually is.

2.3 Low and Stale Prices

To deal with price quality issues that cannot be resolved by the missing-values treatment described above, we add two additional filters. The first filter treats prices below \$0.50 as missing and replaces them with `NaN`. Securities trading at such low levels (Penny Stocks) can generate large or even infinite return from small price movements, keeping them in the sample would distort the expected return and the covariance estimates used in the optimization. This filter is applied before computing the returns so that the observations are not carried into the subsequent forward-fill procedure. The second filter targets those securities that have low trading volume so their prices remain essentially unchanged over long periods. For each firm and each formation year Y, the fraction of months with a return of exactly zero is calculated over the full 120-month estimation window $[Y-9, Y]$, regardless of how many of those months have valid price observations, this is important because if the zero-return share were computed only over the valid months with observations, illiquidity would be understated for firms with many missing prices, since missing values and zero returns are two different signs of the same underlying problem. By always using 120 months as the denominator, the filter is applied consistently across all the firms, any firms where this fraction exceeds 50% is classified as illiquid and excluded from the investment set for that year. These two filters remove securities whose price behavior would distort the covariance matrix used in the optimization.

2.4 Carbon Data Preparation

Scope 1 and Scope 2 greenhouse gas emissions at the Firm-level are taken from two separate Datastream files, each file covers 27 years of annual observations. Scope 1 captures the direct emissions from operations owned or controlled by the firm, Scope 2 captures indirect emissions from purchased electricity and heat. After loading, both series are restricted to the Emerging Markets universe identified in the price data, and the same standardization and transposition procedure described in Section 2.1 is applied. Annual reporting gaps, years in which a firm does not submit emissions data are handled with a forward-fill procedure similar to the one used for price data. When a firm does not report emissions in a given year the most recent available value is carried forward to the following years. This treatment is only applied to gaps that appear in the middle or the end of the available emissions history. When emissions data are missing at the beginning of the sample, no correction is applied: the firm cannot enter the investment set until carbon data becomes available, for the same reason that early price gaps are left untreated. Filling early missing emissions values by going backwards would not represent correctly the firm's environmental profile in a period where no information is available.

The revenue data needed for the carbon intensity calculations in Section 4.1 are reported by Datastream in thousands of US dollars and are converted to millions of US dollars before any ratio is computed. This conversion is necessary to express carbon intensity in the standard unit of tonnes of CO_2 equivalent per million dollars of revenue (tCO_2e / \$M rev.), as required by the WACI and CF formulas introduced in Part II. Firms for which no carbon data are available in any year of the sample are permanently removed from all datasets and cannot enter the investment set under any circumstances.

2.5 Dynamic Investment Set Construction

For each formation year Y, the investment set for the following holding year $Y+1$ is rebuilt using a rolling 10-year estimation window, spanning from January $Y-9$ to December Y, for a total of $\tau = 120$ monthly observations. This rolling structure ensures that the portfolio always uses the most recent available information and that the covariance matrix is never based on data from periods that are no longer relevant.

A firm enters the investment set for holding year $Y+1$ if and only if all five of the following conditions are met at the end of year Y:

(i) it belongs to the Emerging Markets region, as defined by the Datastream regional classification used to build the initial universe;

(ii) it has a valid price above \$0.50 at the end of year Y, so that the portfolio does not include penny stocks;

(iii) it has at least 60 non-missing monthly returns over the 120-month estimation window — a more conservative threshold than the 36-month minimum suggested in the project guidelines, chosen to produce more reliable covariance matrix estimates;

(iv) the proportion of zero-return months over the full 120-month window does not exceed 50%;

(v) both Scope 1 and Scope 2 emissions data are available for year Y, ensuring consistency between the financial optimization and the carbon constraints.

Condition (iii): Raising the minimum return history from 36 to 60 months reduces the number of eligible firms, especially in the early years of the sample, but it significantly improves the quality of the covariance estimates. With fewer than 60 valid observations, the sample covariance matrix of a large set of assets becomes very unstable and tends to produce extreme portfolio weights. The stricter threshold is therefore a deliberate choice to balance the size of the investment set against the reliability of the estimates.
Condition (v): This condition is applied from the beginning of the sample to ensure consistency between Part I and Part II. Restricting the investment set to firms with available carbon data means that all portfolios are built from the same universe, whether they include a carbon constraint or not. This way, when comparing strategies, any difference in results can be directly linked to how the portfolio was constructed, rather than to a different set of underlying firms.

Table 1 reports the number of eligible firms per holding year over the full out-of-sample period from 2014 to 2025.

Table 1: Eligible firms per holding year (10-year rolling window, 2014–2025)

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Firms	245	266	296	334	380	415	463	503	540	569	578	566

Note: Eligible firms are those satisfying all five criteria described in Section 2.5. The investment set is determined at the end of year Y and implemented during holding year $Y+1$.

Table 1 shows that the eligible universe grows steadily over time, from 245 firms in 2014 to a peak of 578 in 2024, before falling slightly to 566 in 2025. This trend mainly reflects the

gradual adoption of emissions reporting by Emerging Markets firms over the past decade. As more firms start disclosing Scope 1 and Scope 2 data, condition (v) becomes less restrictive for a growing part of the universe. The small drop in 2025 most likely reflects a few firms that delisted or lost continuous data coverage over the required estimation window.

The growth of the investment set has an important consequence for covariance estimation. In the later years of the sample, the number of eligible firms N gets close to and eventually exceeds the length of the estimation window $\tau = 120$. When $N > \tau$, the sample covariance matrix is singular and cannot be inverted directly. To deal with this, missing returns within the estimation window are replaced by zero before the covariance matrix is computed. This approach keeps the $\tau \times \tau$ structure of the calculation intact, but slightly underestimates the volatility of firms with many missing observations. This limitation is discussed in Section 6.

3 Standard Portfolio Allocation

This initial section establishes the conventional benchmark against which the more sustainable allocations will be later compared. This project considers two of the most standard asset allocation over an out of sample period from January 2014 to December 2025. The first portfolio is the so called *minimum-variance portfolio*, where the goal is to choose an assets' combination that minimizes portfolio variance. The second portfolio considered is the *value-weighted portfolio*, where assets' weight are based on the market capitalization, meaning bigger companies hold heavier weights. The purpose of this section is to give a first glance of financial performances of our universe but also it provides a foundation, empirical and methodological, that is required for the next sections of this project. Any evaluation with constraints has to be anchored in a clear and well defined environment of the underlying financial world.

3.1 Methodology

Data cleaning and return construction

Raw data come from the Datastream's platform and therefore need preliminary cleaning before computation. Main corrections are due to a Datastream error flag, `$$ER`, non numeric value are dropped and every dataset is then transposed, in order to have rows indexed by the dates and columns defined by firms identifiers. Every month, indices' price are converted into discrete returns using this formula:

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1, \tag{1}$$

where $P_{i,t}$ denotes the total return index of firm i in month t.

Missing observations deserve particular attention, since their treatment can effectively affect the return distribution and also the covariance matrix of the considered universe. Missing values between two dates are forward-filled in order to connect eventual temporary disruptions. In the case where a firms disappears from the list before the end of the sample, a return of -100% is assigned to that firm and all subsequent months are kept as missing. This practice is used to preserve the economic effect of a delisted or defaulted firm without losing the estimated volatility in future computations. Moreover, prices below 0.5 USD are treated as missing values in order to avoid unwanted jumps. All these cleaning procedures allow to preserve a realistic picture of firm exits the list and all other kinds of data imperfections rather than imposing more rigorous rules that would lead to a smoother investment set.

Dynamic investment universe

To build a dynamic investment universe, every year the project uses a rolling 10-year estimation window to rebuild a new set. Concretely, a portfolio build at the end of year Y and implemented at time $Y+1$, the study uses an estimation windows that goes from January $Y-9$ to December Y. This way, $\tau = 120$ monthly observations are implemented whenever a considered asset is available. For a firm to enters the investments set for holding year $Y+1$, it has to satisfies a list of conditions at the end of year Y. Conditions to satisfy are as follows for every firm X:

1. the firm belongs to the Emerging Markets universe;

2. a valid price at the portfolio formation date;
3. at least 60 valid monthly return observations over the preceding 10 years;
4. the fraction of zero monthly returns over the estimation window does not exceed 50%;
5. both Scope 1 and Scope 2 emissions are available for year Y.

Condition number 5 is enforced to have a consistency between a proper financial allocation and the sustainability constraints used in the second part of this project. For the evaluation period, computations shows that the number of eligible firms starts at 245 in 2014 and ends with 566 firms in 2025. This increase is likely related to a progressive implementation of new emissions rules adopted by firms and data availability over time. That said, project's universe is dynamic rather than fixed, allowing a better understanding of new rules implementations. In this setting, such methodologically approach seems appropriate because a static universe would potentially exclude portions of firms' sample. Without the use of the before listed conditions, an additional analysis would have been required to manage incomplete environmental information.

Expected returns and covariance matrix

For the minimum-variance optimized portfolio, only the covariance matrix is required. Nonetheless, the project also computes the expected returns in a way to have a more complete moment estimation framework. For each holding year considered, a firm missing return is set to zero in order to maintain a balanced environment of $\tau = 120$ months. However, this methodological choice should be interpreted as a simplification, since replacing missing returns with a value of zero automatically reduces the estimated volatility and covariance of the sample. Finally, the expected return vector is calculated as follows:

$$\hat{\mu}_Y = \frac{1}{\tau} \sum_{k=0}^{\tau-1} R_{t-k}, \tag{2}$$

where t denotes the final month of year Y and k is the time index over τ monthly observations.

Then, the covariance matrix is computed using the same rolling window's approach of τ monthly observations. For each month in the estimation window, the asset returns' vector is then centered by subtracting $\hat{\mu}_Y$ (2). Empirically, the covariance matrix Σ_Y is calculated as an average of outer products of these centered return vectors, more precisely as

$$\Sigma_Y = \frac{1}{\tau} \sum_{k=0}^{\tau-1} \left(R_{t-k} - \hat{\mu}_Y\right) \left(R_{t-k} - \hat{\mu}_Y\right)^\top. \tag{3}$$

The use of the divisor τ, and not $\tau - 1$, maintain the formulation approach used in this analysis. This estimator is directly calculated through a simple matrix multiplication. This computation ensures that this estimation is constructed using only available information, without the need to introduce additional technique like the shrinkage method or other adjustment.

Minimum-variance portfolio

To obtain the weight of the minimum-variance portfolio, every year Y the project solves the following minimizing problem

$$\min_{\alpha_Y} \quad \alpha_Y^\top \Sigma_Y \alpha_Y \tag{4}$$

subject to these two constraints

$$\alpha_Y^\top e = 1, \qquad \alpha_{i,Y} \geq 0 \quad \forall i, \tag{5}$$

where e is a vector of ones which ensures that the analysis obtains a fully invested portfolio, hence the weights' sum is equal to one. The second constraint is there to prevent short selling, characterized by a negative weight. This optimization problem is solved using the Sequential Least Squares Programming (SLSQP), that allows an objective function to be solved subject to a variety of constraints, like equality/inequality constraint. The project rounds the resulting weights up to six decimals and normalize them to ensure that the budget constraint is satisfied. This computation does not try to find the securities that will lead to the highest expected returns. Instead, in this section the analysis wants to find the portfolio allocation with the lowest variance possible. Using only the constraints shown in Equation (5), the project does not implement an upper bound for every asset's weight. This is important for the final interpretation of results, because with this structure the composition of the minimum-variance portfolio is driven almost only by the estimated covariance matrix (3).

For this analysis, portfolios are rebalanced once per year. The following year, the portfolio returns is calculated every month as

$$R_{p,t+k}^{(mv)} = \alpha_{t+k-1}^\top R_{t+k}, \qquad k = 1, \ldots, 12, \tag{6}$$

where also portfolio weights change according to

$$\alpha_{i,t+k} = \alpha_{i,t+k-1} \frac{1 + R_{i,t+k}}{1 + R_{p,t+k}^{(mv)}}. \tag{7}$$

This is equal to an annual rebalancing with an additional buy and hold weight drift during the considered year. Repeating these calculations over the entire sample, the analysis obtains a series of 144 monthly portfolio returns.

Value-weighted benchmark

The second portfolio calculated in this section is the *value-weighted portfolio*, and it uses the same rules that define the minimum-variance universe. It is crucial to use the same setup to obtain meaningful values that are then comparable with the portfolio obtained before. Monthly returns are calculated as

$$R_{p,t}^{(vw)} = \sum_{i=1}^{N_t} w_{i,t-1} R_{i,t}, \tag{8}$$

where assets' weights are proportional to the size of each firm in the considered universe:

$$w_{i,t-1} = \frac{Cap_{i,t-1}}{\sum_{j=1}^{N_t} Cap_{j,t-1}}. \tag{9}$$

Lagged market capitalization $(t-1)$ are used to prevent look-ahead bias. This is done in order to use only firm's information available at the end of the previous month, which is then used to sum the returns realized in month t. This benchmark has to be interpreted has a passive market benchmark that uses the same eligible universe as the optimized strategy of the minimum-variance portfolio.

Performance metrics

To be able to understand how portfolios evolved through time, the project then estimates certain performance metrics. The main measure of performance is defined as the cumulative return of the portfolio over the 144 months period. This value is defined by

$$R_p^{cum} = \prod_{t=1}^{T}(1 + R_{p,t}) - 1, \tag{10}$$

and in the same style, the annualized cumulative return is computed as

$$R_p^{ann,cum} = \left(\prod_{t=1}^{T}(1 + R_{p,t})\right)^{12/T} - 1. \tag{11}$$

The project computes also the arithmetic average return, and respectively the annualized volatility of portfolios

$$\bar{\mu}_p^{ann} = 12\,\bar{R}_p, \tag{12}$$

and

$$\sigma_p^{ann} = \sqrt{12}\,\sigma(R_{p,t}). \tag{13}$$

Finally, to compute the portfolio's risk-adjusted return, or Sharpe Ratio, the analysis uses the following formula:

$$SR_p = \frac{12(\bar{R}_p - \bar{R}_f)}{\sqrt{12}\,\sigma(R_{p,t})}. \tag{14}$$

These metrics are calculated following a standard empirical asset-pricing practice (Estrada, 2010), where cumulative returns are computed in geometric terms, whereas average values and volatility are annualized from arithmetic monthly moments. Finding these metrics is crucial because it allows the project to reports different scale of realized performance. On one hand, the computations give us a terminal wealth accumulation and in the other hand an average of the monthly payoffs. This distinction is relevant when comparing a more defensive strategy as the minimum-variance with a more standard and exposed benchmark strategy.

3.2 Results

This section explores the results obtained in this chapter. Table 2 summaries and confronts the performances of the *minimum-variance* and *value-weighted* portfolios.

Table 2: Out-of-sample performance comparison, January 2014–December 2025

Metric	minimum-variance	Value-Weighted
Total cumulative return (%)	109.53	131.34
Annualized average return (%)	6.72	8.22
Annualized cumulative return (%)	6.36	7.24
Annualized volatility (%)	10.48	15.57
Sharpe ratio	0.475	0.416
Minimum monthly return (%)	-7.59	-16.71
Maximum monthly return (%)	12.13	13.41

Minimum-variance portfolio

For the period 2014-2025, the *minimum-variance portfolio* delivers a cumulative return of 109.53%, equivalent to a return of 6.72% in arithmetic value and a 6.36% rate expressed in geometric growth. The analysis shows that this strategy achieves an annualized volatility of only 10.48%, which is below the benchmark level of the *value-weighted portfolio*.The effective Sharpe ratio of 0.475 indicates that this strategy converts the risk-adjusted return efficiently, even if this approach does not looks for a maximization of the expected returns. With these results, the report achieves a first important result: a strategy based on the minimization of the variance can produce substantial returns and at the same time enhances the return per risk. This strategy also has a return profile that is relatively defensive. Table 2 shows how the lowest monthly loss stops at -7.59%, while the higher monthly return reaches a value of 12.13%. These results are consistent with the objective function stated in Equation (4). With a minimum-variance strategy, an investor sacrifices part of highest returns in exchange for a more flat but stable return path, with an additional protection against lower returns. In practice, a portfolio allocation that follows this strategy behaves as a conservative assets allocation. This portfolio allocation does not seek for the highest expected returns possible but rather it tries to combine assets with an optimized covariance structure that minimizes the aggregate portfolio's variance. An investor that prioritizes a stability of returns will be attracted to this particular strategy. That is because the portfolio will benefits from the up movements of the market and at the same time will be protected from the downside of returns.

Value-weighted benchmark

Looking at the *value-weighted* strategy, the project reports an higher total return of 131.34%, that translates to higher annualized average return and annualized return, respectively of 8.22% and 7.24%. These values highlights the fact that a passive allocation is able to capture more upside market phases. But this higher return comes with a cost, that is an higher risk. This strategy shows an annualized volatility of 15.57%, that is about +5% higher than the minimum-variance portfolio. The analysis concludes that such allocation dominates when the investor prefer an higher wealth creation, but drawback with its risk level.

Minimum monthly return is also affected, with a lowest value of -16.71%, which is more than twice the value obtained with the minimized variance portfolio. Although a significant Sharpe ratio of 0.416, the benchmark strategy does not give enough additional return in order to compensate the increased volatility carried by this type of allocation.

These results are consistent with an economic point of view of capitalization weighting. With such a strategy, larger weights are assigned to firms with an higher market capitalization, that translates to a larger exposure to stocks whose prices have risen more; therefore carrying more pro-cyclical movements of the market (Wurgler, 2010). On one hand, when market conditions are positive, this strategy delivers stronger wealth creation. On the other hand, during not favorable or recession periods drawdowns are amplified, meaning a bigger return dispersion. To conclude the analysis of the *value-weighted portfolio*, it is necessary to say that such benchmark is used mainly to understand the natural and passive aggregate cycle of the considered assets' universe.

Comparative assessment

This first part of the analysis reveals a clear difference between a return's maximization and a risk reduction strategy. The *value-weighted* benchmark excels in absolute returns, while the *minimum-variance* strategy is better when taking into account risk-adjusted terms. This project uses the same eligible assets' universe, so the differences in the results can be explained only when looking at the portfolio construction rather than to a difference in the sample used. This initial comparison is then useful only in a structural sense, since it only explains how the portfolio behavior is influenced by a passive market exposure and on how much this trend can change when imposing a risk-minimization constraint.

The trade-off between these two different strategies is better summarized Figure 1. The *value-weighted* benchmark exhibits an higher terminal wealth in the considered period. But this better performance comes with a less smooth return path, as shown for example with a deepest peak in 2020 or with considerable bigger return in 2021-2022. A more regular trend can be identified for the *minimum-variance* portfolio. This strategy avoids most sever market fall, which is exactly what an investor who choose this type of allocation look for: lower return in exchange of a smaller covariance. Figure 1 captures the fundamental differences between a strategy that accepts to follow market's fluctuations versus a strategy that actively tries to modify portfolio weights in order to reduce market exposure.

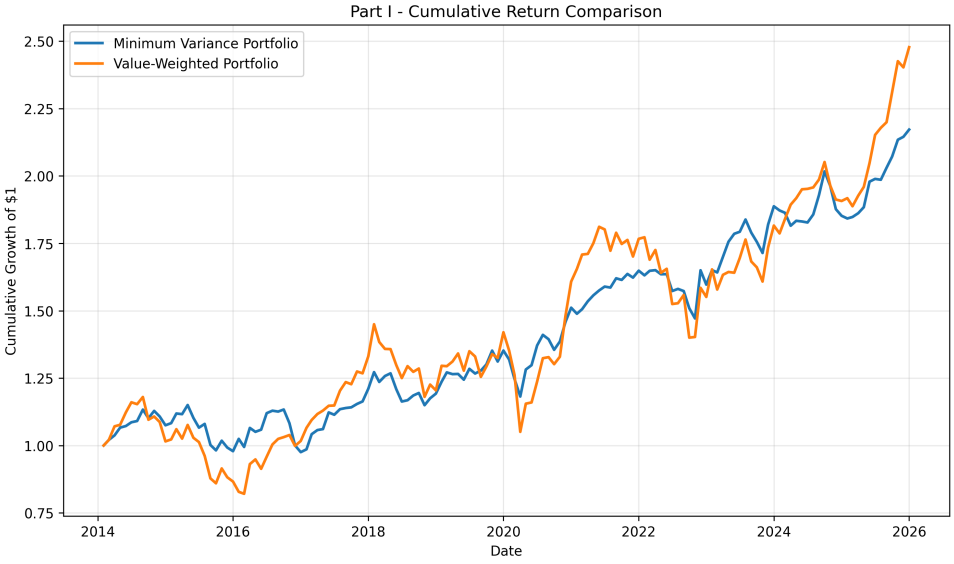


Figure 1: Cumulative return comparison

The graph in Figure 2 better shows the volatility argument. The *value-weighted* portfolio exhibits more volatility and sharper spikes can be seen during specific subperiods. As expected, a lower realized risk level is maintained from the *minimum-variance* strategy all along the sample period. This evidence supports that the lowest volatility reported in Table 2 does not come from few isolated cases, but rather shows a persistent characteristic of this allocation. That said, a *minimum-variance* portfolio is safer as it protects the investor from market's fluctuations.

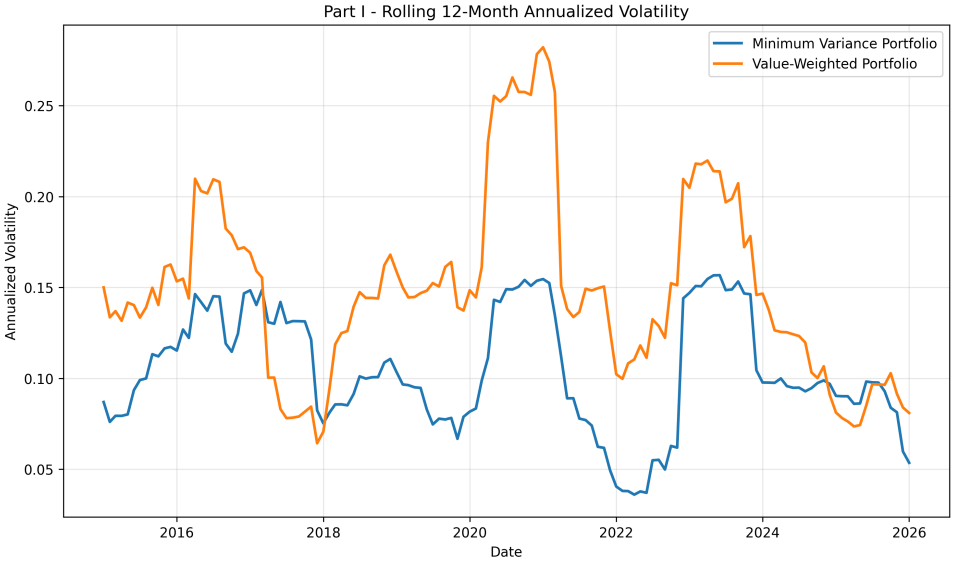


Figure 2: Rolling 12-month annualized volatility

Figure 3 gives us another interesting comparison between the *minimum-variance* portfolio and the *value-weighted* benchmark as it shows how the two strategies allocate differently capital across countries over time. On the right chart, the *value-weighted* strategy appears to have a more geographically diversified and stable trend, since a country's weight evolves gradually through time across different markets. This result is consistent with the enforced constraint. Portfolio's weights follow the market capitalization structure of the eligible universe, so they naturally reflect the geographical composition of the analysis's assets universe. The *minimum-variance* chart, on the left, shows a geographical profile that is much more unstable and concentrated. The allocation profile of this strategy assigns larger weight to a smaller number of countries and presents a less smoother trend. It is worth noting the example of Taiwan (light blue), which across time maintains a significant weight in the portfolio. This results should not be interpreted as that Taiwan carries the lowest standalone variance across countries, since weights are determined by the full covariance matrix, therefore depending on each country's marginal contribution to risk (Clarke et al., 2011).From an economic point of view, this persistent presence of Taiwan's equities are plausibly given by the structure of its equity market, which is highly specialized in the semiconductor industry. Taiwan's semiconductor firms benefit from technological expertise, vertical specialization, cluster effects and advanced production processes (Wang & Chiu, 2014). These characteristics of the Taiwanese economy may explain why Taiwan securities participate in reducing the risk exposure of this strategy. In other words, results obtained in Table 2 and in Figure 2 are partly related to a different geographical reallocation, with Taiwan's securities playing an important role on minimizing the variance.

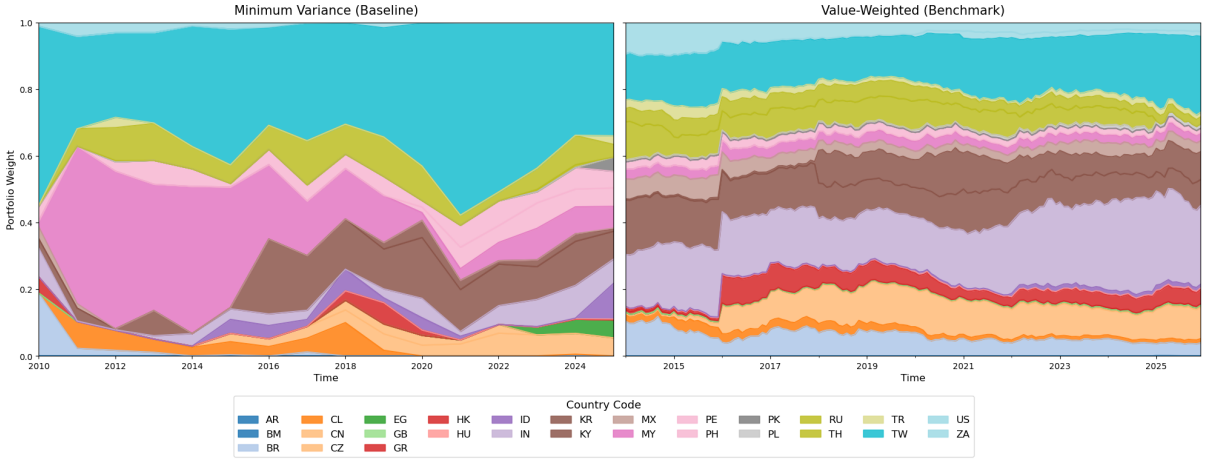


Figure 3: Country allocation of the minimum-variance and value-weighted portfolios

To conclude, this simple analysis shows more than a simple return's difference between these two strategies. The *value-weighted* portfolio performs better in absolute terms, but carries a larger instability and materially larger loss. The *minimum-variance* portfolio is less aggressive, but offers a better protection against volatility and at the end delivers a higher Sharpe ratio. This last strategy constitutes a preferred benchmark for the sustainability constraints that will be later introduced in this report.

4 Portfolio Allocation with Carbon Emission Reduction

In this part the baseline allocation framework is integrating carbon emissions into the portfolio creation. Following the project instructions, first we compute portfolios benchmark's environmental profile by using the Weighted-Average Carbon Intensity (WACI) and Carbon Footprint (CF), then we implement two decarbonization methods. The first tactic is designed for an active investor and sets a 50% reduction in portfolio-related emissions relative to the minimum-variance portfolio. The second tactic follows the approach of a passive investor and minimizes tracking error with respect to the value-weighted benchmark while imposing the same 50% carbon-footprint reduction.

4.1 Carbon Metrics

Methodology

To assess the emission profile of the portfolio, we calculated two standard carbon metrics based on firm-level Scope 1 and Scope 2 CO_2 emissions.

The first metric is firm-level *carbon intensity*, expressed as tonnes of CO_2 corresponding per million U.S. dollars of revenue:

$$\mathrm{CI}_{i,Y} = \frac{E^{(1)}_{i,Y} + E^{(2)}_{i,Y}}{\mathrm{Rev}_{i,Y}}, \tag{15}$$

Where $E^{(1)}_{i,Y}$ and $E^{(2)}_{i,Y}$ denote Scope 1 and Scope 2 carbon output, correspondingly, and $\mathrm{Rev}_{i,Y}$ is the yearly revenue measured in millions of U.S. dollars. Since Datastream shows revenues in thousands of U.S. dollars, revenues are divided by 1,000 before calculating the ratio.

At the portfolio level, we first determine the *Weighted-Average Carbon Intensity*:

$$\mathrm{WACI}^{(p)}_{Y} = \sum_{i=1}^{N} \alpha_{i,Y}\, \mathrm{CI}_{i,Y}, \tag{16}$$

where $\alpha_{i,Y}$ is the portfolio allocations assigned to stock i at the yearly rebalancing date.

The second climate-related measure is the *Carbon Footprint*, which shows the amount of firm-level carbon emissions financed by the investor per million U.S. dollars allocated:

$$\mathrm{CF}^{(p)}_{Y} = \frac{1}{V_Y} \sum_{i=1}^{N} o_{i,Y} E_{i,Y}, \qquad o_{i,Y} = \frac{\alpha_{i,Y} V_Y}{\mathrm{Cap}_{i,Y}}, \tag{17}$$

where $E_{i,Y}$ is aggregate firm emission, $\mathrm{Cap}_{i,Y}$ is market capitalization at year-end, $o_{i,Y}$ is the inferred ownership stake of the firm represented in the portfolio, and V_Y represents the portfolio value. The portfolio is established with starting wealth $V_{2013} = \$1{,}000{,}000$.

For the value-weighted benchmark, the emission footprint is equivalent to:

$$\mathrm{CF}^{(\mathrm{vw})}_{Y} = \frac{\sum_i E_{i,Y}}{\sum_i \mathrm{Cap}_{i,Y}}, \tag{18}$$

because benchmark weights are scaled by to market capitalisation.

These two metrics are related. WACI measures the emission intensity of the companies held in the portfolio, while CF captures the emissions financially attributed to the investor based on ownership.

Results

Table 3 shows the yearly carbon metrics of the unconstrained minimum-variance portfolio $P_{\text{oos}}^{(\text{mv})}$ and of the value-weighted benchmark $P^{(\text{vw})}$ over the 2014–2025 out-of-sample period.

Table 3: Annual carbon metrics of $P_{\text{oos}}^{(\text{mv})}$ and $P^{(\text{vw})}$ (2014–2025)

	$P_{\text{oos}}^{(\text{mv})}$		$P^{(\text{vw})}$	
Year	**CF** ($\text{tCO}_2\text{e}/\$\text{M}$)	**WACI** ($\text{tCO}_2\text{e}/\$\text{M rev.}$)	**CF** ($\text{tCO}_2\text{e}/\$\text{M}$)	**WACI** ($\text{tCO}_2\text{e}/\$\text{M rev.}$)
2014	156.90	294.28	463.61	572.01
2015	236.64	676.09	582.29	631.97
2016	188.06	486.51	463.96	516.44
2017	271.87	421.26	707.53	627.97
2018	136.37	371.64	664.46	702.10
2019	375.00	885.79	590.58	484.91
2020	275.01	628.22	690.26	587.85
2021	321.84	438.86	673.99	416.44
2022	400.57	669.59	589.68	423.10
2023	475.67	374.96	571.08	391.10
2024	385.91	382.86	651.47	356.44
2025	648.21	484.86	582.31	395.49

The analysis indicates a clear environmental gap across the two baseline portfolios. In most years of the sample period, the value-weighted benchmark displays a significantly larger carbon emission than the minimum-variance portfolio. Economically, this outcome is intuitive. The benchmark assigns more capital to the largest companies in the investment universe, including utilities, energy, and materials companies, which are frequently high-emission firms. By contrast, the minimum-variance portfolio is chosen according to covariance properties and naturally tilts towards less volatile companies, which during the evaluation period also tend to be associated with lower financed emissions.

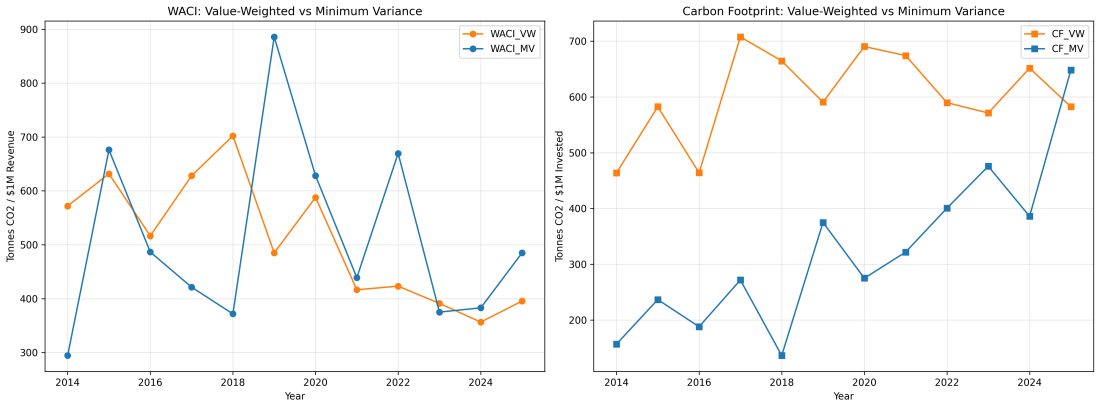


Figure 4: Annual WACI and carbon footprint of the two baseline portfolios, $P_{\text{oos}}^{(\text{mv})}$ and $P^{(\text{vw})}$, over 2014–2025.

The same finding broadly appears from the WACI comparison. Although the relative positioning is not the same on an annual basis, the minimum-variance portfolio overall exhibits lower emissions intensity than the value-weighted benchmark. This indicates that the baseline risk-minimization strategy already integrates an underlying environmental tilt, even prior to any clear constraint is introduced.

To better understand which companies drive the carbon intensity of the portfolios, Tables 4 and 5 present the top ten average contributors to WACI over the sample period.

Table 4: Top 10 average WACI contributors in $P^{(\mathrm{vw})}$

Rank	Company	ISIN	Avg. Contribution
1	NTPC	INE733E01010	69.14
2	ULTRATECH CEMENT	INE481G01011	31.52
3	GAZPROM	RU0007661625	23.50
4	SASOL	ZAE000006896	18.41
5	IOI CORPORATION	MYL1961OO001	11.81
6	TAIWAN SEMICON. MNFG.	TW0002330008	10.66
7	OC ROSNEFT	RU000A0J2Q06	9.70
8	TENAGA NASIONAL	MYL5347OO009	9.57
9	RELIANCE INDUSTRIES	INE002A01018	9.45
10	AMBUJA CEMENTS	INE079A01024	8.83

Table 5: Top 10 average WACI contributors in $P_{\mathrm{oos}}^{(\mathrm{mv})}$

Rank	Company	ISIN	Avg. Contribution
1	SEMEN INDONESIA (PERSERO)	ID1000106800	88.60
2	ENGIE ENERGIA PERU	PEP702101002	58.25
3	ORYGEN PERU	PEP700511004	55.16
4	RATCH GROUP PCL	TH0637010Y00	50.52
5	ELECTRICITY GENERATING	TH0465010005	45.77
6	COSCO SHIP.EN.TRSP. 'H'	CNE1000002S8	21.89
7	CHUNGHWA TELECOM	TW0002412004	19.35
8	PETRONAS GAS	MYL6033OO004	17.86
9	COLBUN MACHICURA	CLP3615W1037	16.75
10	FAUJI FERTILIZER COMPANY	PK0053401011	12.37

The benchmark's greatest contributors are mainly large carbon-intensive firms, such as Indian power producers, cement manufacturers, and oil and gas companies. On the other hand, the minimum- variance portfolio's leading drivers are more focused on utilities and defensive industrial firms from South-East Asia and Latin America. This disparity is consistent with the differing aims of the two strategies: the benchmark mirrors market capitalization, whereas the minimum-variance portfolio reflects statistical risk features.

4.2 Carbon-Constrained minimum-variance Portfolio

Methodology

We now apply an explicit carbon-footprint decline on the minimum-variance portfolio. For every formation year Y, the investor selects a long-only portfolio that minimizes

forecasted variance subject to a carbon footprint cap equivalent to 50% of the carbon footprint of the unconstrained minimum-variance portfolio:

$$\min_{\alpha_Y} \alpha_Y^\top \Sigma_Y \alpha_Y \quad \text{s.t.} \quad \mathrm{CF}_Y^{(p)} \leq 0.5 \times \mathrm{CF}_Y^{\left(P_{\text{oos}}^{(\text{mv})}\right)}, \quad \alpha_Y^\top e = 1, \quad \alpha_{i,Y} \geq 0 \ \forall i. \tag{19}$$

The baseline carbon footprint $\mathrm{CF}_Y^{\left(P_{\text{oos}}^{(\text{mv})}\right)}$ is re-evaluated at every annual rebalancing date from the unconstrained minimum-variance weights. We denote the resulting constrained portfolio by $P_{\text{oos}}^{(\text{mv})}(0.5)$. From an economic perspective, this issue measures the cost of obliging a risk-minimizing investor to reduce financed emissions by half to the original active portfolio.

Results

Table 6 confirms that the carbon-footprint target is met in every year of the sample.

Table 6: Carbon-footprint enforcement for $P_{\text{oos}}^{(\text{mv})}(0.5)$

Year	Baseline CF (MV)	Target CF (50%)	Actual CF
2014	156.90	78.45	78.45
2015	236.64	118.32	118.32
2016	188.06	94.03	94.03
2017	271.87	135.93	135.93
2018	136.37	68.19	68.19
2019	375.00	187.50	187.50
2020	275.01	137.50	137.50
2021	321.84	160.92	160.92
2022	400.57	200.28	200.28
2023	475.67	237.84	237.84
2024	385.91	192.95	192.95
2025	648.21	324.11	324.11

Table 7 compares the financial performance of the unconstrained and carbon-constrained minimum-variance portfolios.

Table 7: Performance comparison: $P_{\text{oos}}^{(\text{mv})}$ vs. $P_{\text{oos}}^{(\text{mv})}(0.5)$ (2014–2025)

Metric	$P_{\text{oos}}^{(\text{mv})}$	$P_{\text{oos}}^{(\text{mv})}(0.5)$
Annualized Average Return	6.72%	6.52%
Annualized Volatility	10.48%	10.03%
Annualized Cumulative Return	6.36%	6.19%
Sharpe Ratio	0.47	0.48
Minimum Monthly Return	-7.59%	-6.76%
Maximum Monthly Return	12.13%	11.93%

The outcome indicates that a 50% decrease in financed emission can be achieved with only a modest deterioration in financial performance. Annualized average return declines

marginally from 6.72% to 6.52%, while the Sharpe ratio stays almost unchanged, rising slightly from 0.47 to 0.48. Simultaneously, annualized volatility drops from 10.48% to 10.03%. This suggests that some of the omitted high-emissions securities were not only carbon intensive but also relatively risky, therefore the environmental constraint does not necessarily increase overall portfolio volatility.

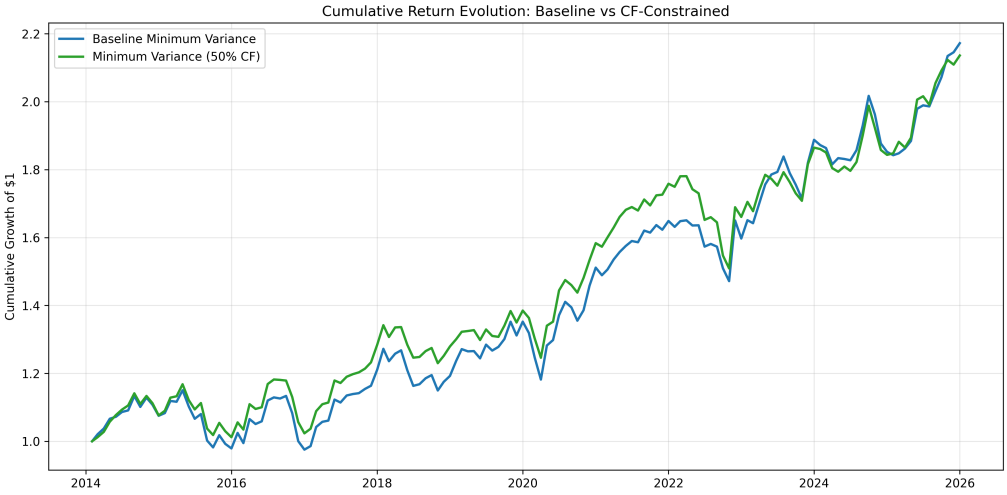


Figure 5: Cumulative returns of $P_{\text{oos}}^{(\text{mv})}$ and $P_{\text{oos}}^{(\text{mv})}(0.5)$.

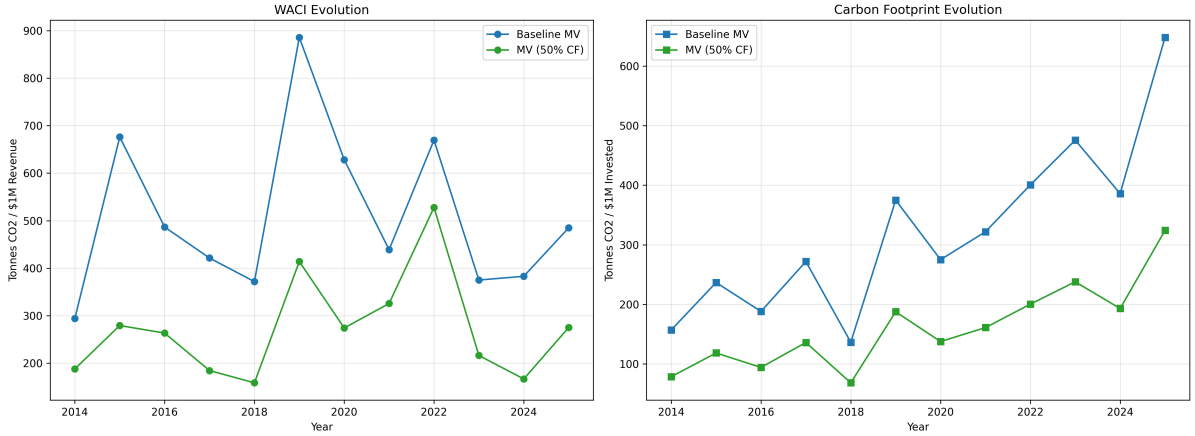


Figure 6: WACI and carbon footprint evolution of $P_{\text{oos}}^{(\text{mv})}$ and $P_{\text{oos}}^{(\text{mv})}(0.5)$.

The cumulative-return graph confirms that the constrained portfolio stays near to the original minimum-variance approach over the whole out-of-sample period, while the carbon-evolution plot shows that both WACI and CF are significantly reduced. The decarbonized active portfolio therefore keeps most of the defensive characteristics of the original minimum-variance allocation with only a slight decline in annualized return and generally unchanged risk-adjusted performance.

To gain a clearer understanding of how the carbon constraint changes the structure of the portfolio, Tables 8 and 9 summarize the main removed firms and reallocation generated by the optimization.

Table 8: Top 5 firms most often excluded from $P_{\text{oos}}^{(\text{mv})}(0.5)$

Company	ISIN	Years Excluded
TAIWAN MOBILE	TW0003045001	5
NESTLE (MALAYSIA)	MYL4707OO005	4
FAR EASTONE TELECOM.	TW0004904008	4
ENGIE ENERGIA PERU	PEP702101002	3
POWER GRID CORPORATION OF INDIA	INE752E01010	3

Table 9: Largest average weight shifts in $P_{\text{oos}}^{(\text{mv})}(0.5)$ relative to $P_{\text{oos}}^{(\text{mv})}$

Company	Baseline MV	MV (0.5)	Shift
Most under-weighted positions			
BRIT.AMER.TOB.(MALAYSIA)	14.57%	5.69%	-8.88%
TAIWAN MOBILE	9.29%	2.79%	-6.50%
ADVANCED INFO SER.	3.60%	1.26%	-2.34%
CPFL ENERGIA ON	1.71%	0.00%	-1.71%
TELEKOM MALAYSIA	4.24%	2.62%	-1.63%
Most over-weighted positions			
CHUNGHWA TELECOM	25.45%	31.58%	+6.14%
PPB GROUP	0.36%	0.84%	+0.49%
THAI UNION GROUP	2.65%	3.14%	+0.48%
ASUSTEK COMPUTER	0.09%	0.55%	+0.46%
KLCC PROPERTY HOLDINGS STAPLED UNITS	1.23%	1.67%	+0.44%

Carbon restriction consistently lowers exposure to companies that are attractive from a pure variance point of view but relatively carbon intensive, particularly in utilities and communications. Capital is reallocated toward less carbon-intensive financial and defensive stocks. Thus, for the active investor, decarbonization comes with a limited but noticeable cost in performance and Sharpe ratio, even though the size of the cost stays economically limited.

4.3 Tracking Error Minimization

Methodology

We now analyze the passive-investor problem. Instead of keeping to a minimum total portfolio variance, the goal is to stay as close as possible to the value-weighted benchmark meanwhile enforcing a 50% reduction in financed emissions. For each year Y, we solve the following optimization problem:

$$\min_{\alpha_Y}(\alpha_Y - \alpha_Y^{(\text{vw})})^\top \Sigma_Y (\alpha_Y - \alpha_Y^{(\text{vw})}) \quad \text{s.t.} \quad \text{CF}_Y^{(p)} \leq 0.5 \times \text{CF}_Y^{(\text{vw})}, \quad \alpha_Y^\top e = 1, \quad \alpha_{i,Y} \geq 0 \ \forall i. \tag{20}$$

This optimization setup minimizes ex-ante tracking-error variance compared to the benchmark under full-investment and long-only restrictions. We denote the resulting portfolio by $P_{\text{oos}}^{(\text{vw})}(0.5)$.

The economic interpretation varies from the minimum-variance case. Here the investor does not aim to obtain the global minimum-risk allocation, but instead the smallest possible deviation from the benchmark suitable with a strong decarbonization target.

Results

All annual optimizations converge successfully, and the 50% carbon-footprint constraint is binding in every year of the sample.

Table 10: Carbon-footprint enforcement for $P_{\text{oos}}^{(\text{vw})}(0.5)$

Year	Benchmark CF (VW)	Target CF (50%)	Actual CF	Residual
2014	463.61	231.81	231.81	0.00
2015	582.29	291.15	291.15	0.00
2016	463.96	231.98	231.98	0.00
2017	707.53	353.77	353.77	0.00
2018	664.46	332.23	332.23	0.00
2019	590.58	295.29	295.29	0.00
2020	690.26	345.13	345.13	0.00
2021	673.99	337.00	337.00	0.00
2022	589.68	294.84	294.84	0.00
2023	571.08	285.54	285.54	0.00
2024	651.47	325.73	325.73	0.00
2025	582.31	291.16	291.16	0.00

For the *passive investor*, the results are somewhat different from what could be expected from a benchmark-relative strategy. The benchmark-aware decarbonised portfolio $P_{\text{oos}}^{(\text{vw})}(0.5)$ satisfies the imposed 50% carbon-footprint reduction, while also performing better than the unconstrained value-weighted benchmark. Its Annualized return increases from 8.22% to 9.01%, whereas volatility rises only marginally from 15.57% to 16.10%. As a result, the Sharpe ratio improves from 0.42 to 0.45. Over the sample period, therefore, the carbon constraint does not appear to generate a financial cost. On the contrary, in realised terms, the reduction in carbon footprint is associated with slightly stronger performance.

Table 11: Performance comparison: $P^{(\text{vw})}$ vs. $P_{\text{oos}}^{(\text{vw})}(0.5)$ (2014–2025)

Metric	$P^{(\text{vw})}$	$P_{\text{oos}}^{(\text{vw})}(0.5)$
Annualized Average Return	8.22%	9.01%
Annualized Volatility	15.57%	16.10%
Annualized Cumulative Return	7.24%	8.00%
Sharpe Ratio	0.42	0.45
Minimum Monthly Return	-16.71%	-16.72%
Maximum Monthly Return	13.41%	14.19%
Realised Tracking Error (ann.)	0.00%	1.85%

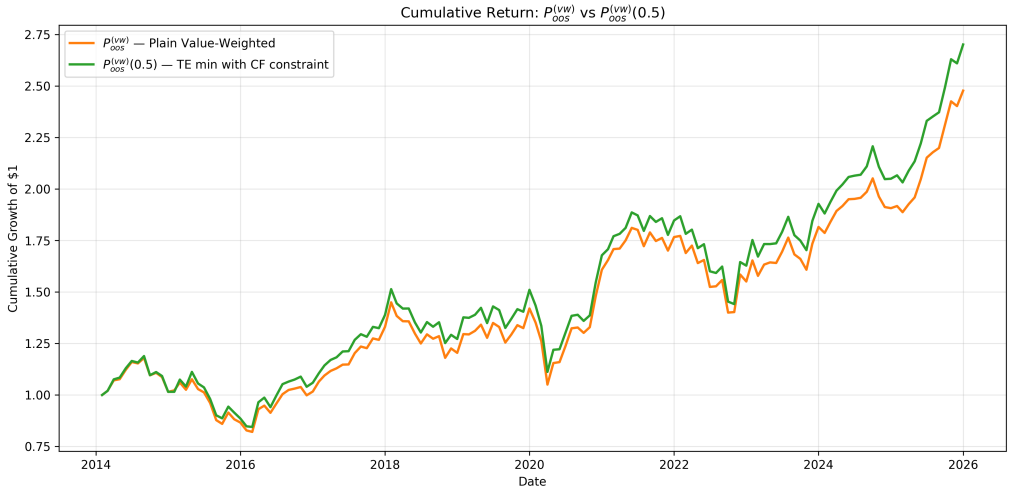


Figure 7: Cumulative returns of $P^{(\mathrm{vw})}$ and $P_{\mathrm{oos}}^{(\mathrm{vw})}(0.5)$.

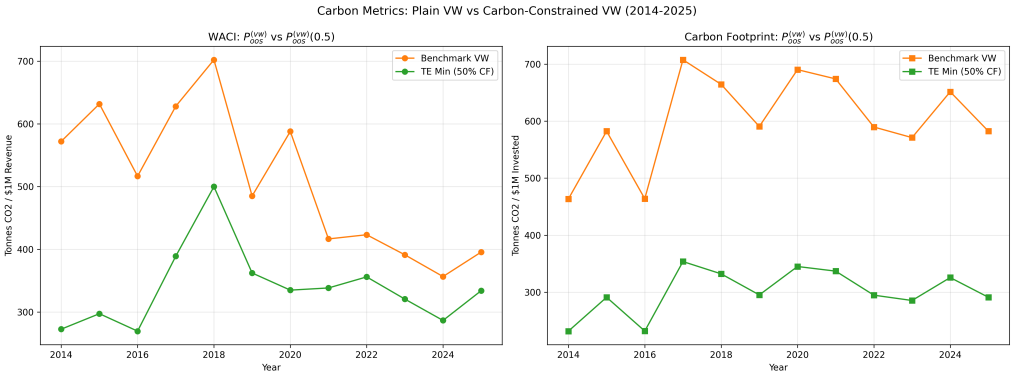


Figure 8: WACI and carbon footprint evolution of $P^{(\mathrm{vw})}$ and $P_{\mathrm{oos}}^{(\mathrm{vw})}(0.5)$.

A possible explanation is that the carbon restriction lowers exposure to high-emission sectors such as energy, utilities, and materials, while reallocating funds toward lower-emission companies that delivered solid performance over the sample period. However, this apparent ''carbon alpha" should be viewed cautiously, as it may indicate sample-specific sector dynamics rather than a durable performance advantage of low-carbon strategies.

4.4 Financial vs. Carbon Performance Trade-off

Table 12 summarises the four portfolios constructed in Part II and illustrates how the financial impact of a carbon constraint depends on the investor's objective.

Two asymmetric results emerge from this comparison.

For the *active investor*, the carbon restriction imposes only a limited return cost. Moving from $P_{\mathrm{oos}}^{(\mathrm{mv})}$ to $P_{\mathrm{oos}}^{(\mathrm{mv})}(0.5)$ reduces annual return slightly from 6.72% to 6.52%, while the Sharpe ratio stays mostly unchanged, moving from 0.47 to 0.48. This suggests that part of the efficiency of the unconstrained minimum-variance allocation came from companies that were empirically attractive from a risk perspective but relatively high in emissions.

Table 12: Summary comparison of the four Part II portfolios (2014–2025)

Metric	$P_{\text{oos}}^{(\text{mv})}$	$P_{\text{oos}}^{(\text{mv})}(0.5)$	$P^{(\text{vw})}$	$P_{\text{oos}}^{(\text{vw})}(0.5)$
Annualized Return	6.72%	6.52%	8.22%	9.01%
Annualized Volatility	10.48%	10.03%	15.57%	16.10%
Sharpe Ratio	0.47	0.48	0.42	0.45
Carbon Footprint Reduction	–	-50%	–	-50%

For the *passive investor*, the result is not exactly what one might expect from a benchmark-relative strategy. The benchmark-aware decarbonized portfolio $P_{\text{oos}}^{(\text{vw})}(0.5)$ meets the 50% carbon-footprint reduction target and also performs better than the unconstrained value-weighted benchmark. Its annualized return increases from 8.22% to 9.01%, while volatility rises only slightly, from 15.57% to 16.10%. As a result, the Sharpe ratio improves from 0.42 to 0.45. Over the sample period, reducing the carbon footprint therefore does not appear to have penalized financial performance.

Overall, Part II shows that the trade-off between financial performance and decarbonization is not universal. In the minimum-variance framework, the carbon constraint slightly lowers return performance, even though it also reduces volatility. In the tracking-error framework, the same emissions constraint leads to a portfolio with higher average return and improved Sharpe ratio, despite a slight increase in volatility and a non-negligible tracking error relative to the benchmark. This distinction is economically important: the cost of decarbonization depends not only on the carbon target itself, but also on the benchmark relative to which the portfolio is designed.

5 Portfolio Allocation with a Net Zero Objective

5.1 Net Zero Portfolio

In the final part of the project the analysis moves from a static decarbonization target to a dynamic Net Zero trajectory. The objective is to create a portfolio that reduces its yearly carbon footprint progressively over a 10-year time framework, while maintaining the investment decisions as close as possible to the market benchmark given by the value-weighted portfolio.

Methodology

The Net Zero portfolio, denoted as $P_{oos}^{(vw)}(NZ)$, is constructed following a tracking error minimization approach similar to one used in Section 4.3, but with a time-varying carbon constraint.

First, a starting point for the progressive minimization problem is required. We evaluated the Carbon Footprint of the Value-Weighted (VW) benchmark at the end of 2013 ($Y_0 = 2013$), and this baseline has been used as the 100% level. Then, from 2014 to 2025, we constructed the portfolio imposing an annual reduction of $\theta = 10\%$.

The carbon emissions reduction constraint for the portfolio in any given year Y is now defined by the following:

$$CF(p)_Y \leq (1-\theta)^{Y-Y_0} \times CF(P^{(vw)})_{Y_0} \tag{21}$$

where Y ranges from 2014 to 2025. Finally, this means that in the final year of the sample, the portfolio is built with a carbon intensity significantly lower (over 70% reduction) than the original 2013 level.

For the practical implementation of the strategy, a constrained optimization problem is solved every year. The objective function is the minimization of the ex-ante Tracking Error variance relative to the Value-Weighted benchmark:

$$\min_{\alpha_Y} \quad (\alpha_Y - \alpha_{Y,vw})' \Sigma_Y (\alpha_Y - \alpha_{Y,vw}) \tag{22}$$

subject to:

- **Full Investment:** $\sum \alpha_{i,Y} = 1$
- **Long-only:** $\alpha_{i,Y} \geq 0$ for all i
- **Net Zero Constraint:** The portfolio Carbon Footprint must stay below the annual target calculated above.

Results

In this section the results are of the Net Zero portfolio ($P_{oos}^{(vw)}(NZ)$) are presented and evaluated against the Value-Weighted (VW) benchmark ($P_{oos}^{(vw)}$). The Net Zero strategy is built to reduce the Carbon Footprint (CF) of the portfolio by 10% every year, with a net-zero objective starting from the 2013 CF level as a base, while minimizing the tracking error relative to the VW benchmark as illustrated in Table 13.

Table 13: Carbon emission and yearly CF reduction

Year	NZ_Target	CF_VW	CF_NZ	Reduction_%
2014	417.25	463.61	417.25	10.00
2015	375.53	582.29	375.53	35.51
2016	337.97	463.96	337.97	27.15
2017	304.18	707.53	304.18	57.01
2018	273.76	664.46	273.76	58.80
2019	246.38	590.58	246.38	58.28
2020	221.74	690.26	221.74	67.88
2021	199.57	673.99	199.57	70.39
2022	179.61	589.68	179.61	69.54
2023	161.65	571.08	161.65	71.69
2024	145.49	651.47	145.49	77.67
2025	130.94	582.31	130.94	77.51

The optimization process has successfully tracked the VW benchmark for the whole sample of years, and kept the portfolio close to the benchmark while precisely enforcing the carbon constraint. As expressed in the optimization diagnostic results, the the realized annualized tracking error for the Net Zero portfolio stood at 1.89% (0.0189). The carbon footprint reduction has been successfully met across all the years under analysis, with the required carbon reduction starting from a 10.0% reduction in 2014 to a 77.5% reduction by 2025 (relative to the 2013 baseline).

Table 14: Performance Metrics: Net Zero vs. Value-Weighted Benchmark

Metric	Net Zero Portfolio	VW Benchmark
Annualized Return	9.16%	8.22%
Annualized Volatility	16.14%	15.57%
Sharpe Ratio	0.4595	0.4157
Minimum Monthly Return	-16.66%	-16.71%
Maximum Monthly Return	14.33%	13.41%
Realized TE (ann.)	1.89%	0,00%

The Net Zero portfolio, despite the strict environmental constraints, slightly outperformed the VW benchmark in terms of financial performance. Table 14 summarizes the annualized risk and return metrics for both portfolios.

The Net Zero portfolio generated an annualized average return of 9.16% compared to the benchmark's 8.22%, with a small increase in annualized volatility (16.14% vs. 15.57%). This results in a higher Sharpe ratio of 0.4595.

Furthermore, the minimum monthly return for the Net Zero strategy (-16.66%) is almost identical to the one of the baseline (-16.71%), then confirming that the decarbonization constraint did not increase risk of major drawdowns.

Portfolio Composition: Exclusions and Weight Shifts

To move to a Net Zero position and meet the annual 10% carbon reduction target it is required a shift in weights, as well as to complete exclusion of some firms in certain years

due to incompatible emissions.

The following tables summarize the most significant adjustments made to the portfolio during the 2014-2025 period.

Table 15: Top 5 Frequently Excluded Firms (2014–2025)

Company Name	Sector	Years Excluded
NTPC	Utilities (Power)	9
CHINA PTL. & CHM. 'H'	Energy (Oil & Gas)	7
CHINA SHENHUA EN.CO. 'H'	Energy/Materials (Coal)	6
CEMEX CPO	Materials (Cement)	6
GRASIM INDUSTRIES	Materials (Conglomerate)	6

Table 16: Weight Shifts: Under-weighted vs Over-weighted Positions

Company	Sector	Shift
NTPC	Utilities	-0.0035
POSCO	Materials	-0.0023
CHINA PTL.	Energy	-0.0017
CEMEX CPO	Materials	-0.0014
GAZPROM	Energy	-0.0014

Company	Sector	Shift
VALE ON	Materials	+0.0011
SBERBANK	Financials	+0.0010
BHARAT HEAVY	Industrials	+0.0009
COCA COLA	Cons. Staples	+0.0009
ASHOK LEY.	Industrials	+0.0008

The exclusion and weighting dynamics reflect a systematic divestment from carbon-intensive activities. The Indian power producer company NTPC is excluded 9 out of 12 years, due to its high reliance on fossil fuels for electricity generation. Other major companies in the Energy and Materials sectors have also been excluded and/or underweighted in many years, being sectors with historically high carbon emissions.

Conversely, the strategy overweights firms with low emissions in order to remain as close as possible to the benchmark. In particular capital is reallocated toward for example Financial, Materials and Industrial companies that have been able to reach a better environmental efficiency.

Geographic Shifts and Capital Reallocation

Alongside the changes in firms weights, there are also some shifts at a macroeconomic level. Indeed, to achieve the Net Zero trajectory, the portfolio's geographic composition is also slightly changed. Figure 9 illustrates the geographic weights of the Net Zero strategy against the standard Value-Weighted benchmark.

As observed in the figure, since the strategy is designed to minimize tracking error against the benchmark, the country weights remain mostly intact, and there are no drastic shifts. This suggests that the necessary carbon reductions are achieved mainly through "within-country" reallocation, divesting from high-emission firms (as shown in Table 16). It can be further noted that the allocations in the Net Zero trajectory show smoother and less volatile changes over time compared to the benchmark.

Interestingly, geographically allocation for the Net Zero portfolio is almost visually identical to from the static 50% Tracking Error reduction portfolio. It can be deduced that the main mechanism used by the optimizer to cut emissions remains consistent, regardless of whether the target is a static 50% cut or a dynamic Net Zero path.

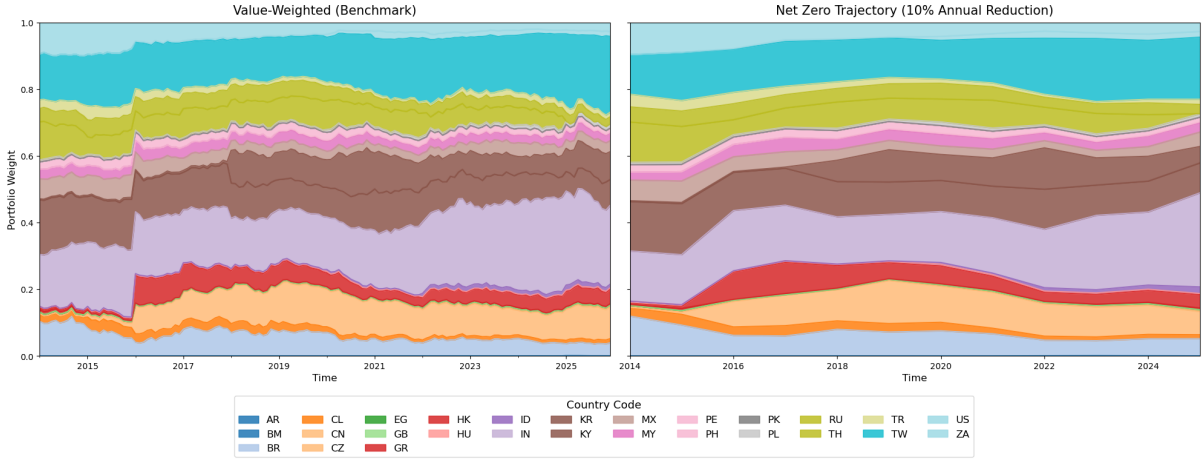


Figure 9: Geographic Allocation: value-weighted vs. Net Zero Trajectory

5.2 Comparison of Decarbonization Strategies

It is important to compare the Net Zero portfolio not only against the Value-Weighted (VW) benchmark, but also against the static 50% Carbon Footprint (CF) reduction portfolio introduced in the previous chapter. This comparison allows us to understand and evaluate the trade-offs between a constant and a progressive carbon constraint.

The main difference between the two different approaches is environmental, as shown in the carbon footprint and carbon intensity trajectories in Figure 10.

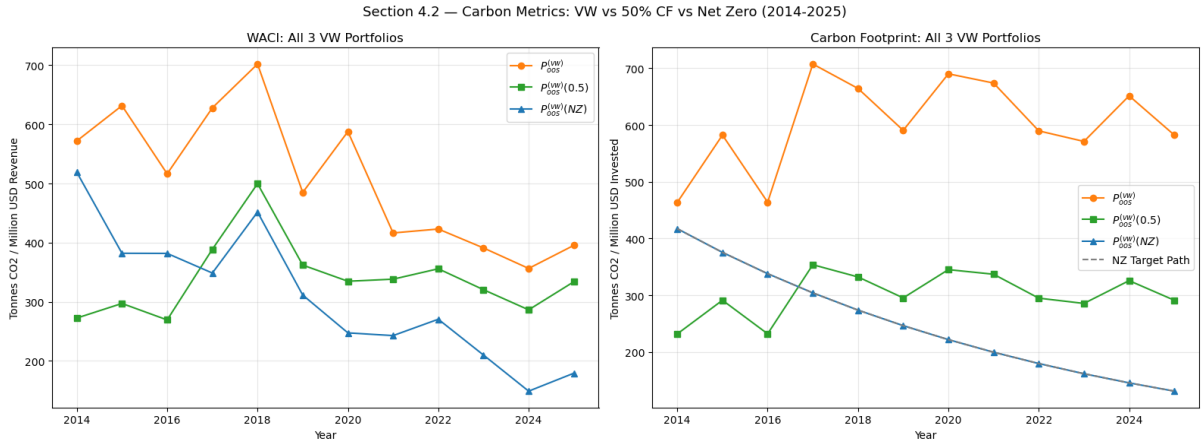


Figure 10: Environmental Trajectories: WACI for the three portfolios over time (left) and Carbon Footprint over time (right).

The 50% CF strategy attains a significant and constant reduction relative to the benchmark, however, it remains a "relative" target. The main difference is that its absolute carbon footprint still fluctuates according to the baseline value. In contrast, the Net Zero strategy only depends on the first year absolute value, and from there it follows a downward path.

One obvious, yet interesting result is that the 50% CF strategy was more aggressive in the initial years (up to 2017) since it directly halves the CF against a 10% progressive reduction. However, as the Net Zero target becomes always lower as it depends on the last year value, it quickly overtakes the static strategy. This result shows the superiority of

the Net Zero approach for long-term climate objectives, as it guarantees that the portfolio moves toward zero over time, regardless of any other portfolio behavior.

From a financial point of view, both decarbonization strategies demonstrated a high degree of efficiency. Figure 11 shows the cumulative returns for all three portfolios.

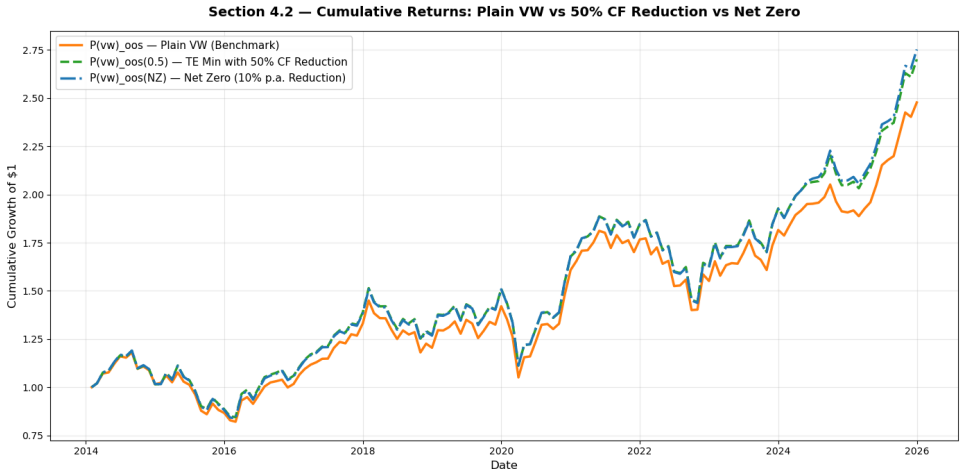


Figure 11: Cumulative Returns Comparison: VW Benchmark, 50% CF Reduction, and Net Zero Strategy.

During the entire sample period, both the 50% CF and the Net Zero portfolios have successfully tracked the VW benchmark, capturing all the main movements and main trends. Interestingly, the Net Zero strategy achieved the highest annualized return (9.16%), followed by the 50% CF strategy (9,01%), both being able to outperform the simple VW portfolio, without any significant increase in risk. Finally, these results suggest that during the last decade, imposing low-carbon constraints (either progressive or static) did not penalize returns, but rather guaranteed a slight increase in performance.

In summary, while a static reduction is an excellent baseline for a sustainable investor, a dynamic and progressive reduction is proven to be a better framework in the longer term.

6 Limitations

The implemented methodology provides a robust framework for analyzing climate-aware portfolios, but there are some limitations to be taken into account.

Data quality and coverage: Carbon emissions data for Emerging Markets firms have some important reporting gaps, in particular before 2010. To maintain a modest investment universe, forward-fill imputation has been used for missing scope 1 and Scope 2 data. However, with this approach some stale values may be introduced for firms that have changed their emissions profiles, potentially leading to incorrect portfolio's carbon constraints optimizations.

Parameter sensitivity: The criteria for defining the investment universe, such as the threshold for stale-prices (excluding firms with zero returns in over 50% of the months) and the minimum valid return requirement (set to 60 months over the 10-year window), are manually chosen. The eligible universe may vary as these parameters are changed, potentially leading to different results in the final portfolios compositions and performances.

Constraint feasibility: The net-zero strategy imposes a progressive constraint that becomes every year smaller in absolute values (falling below 28% of the 2013 baseline in the last year). In the Emerging Markets there may be the risk that the lowest-emissions firm in the eligible universe still exceed this ceiling, forcing the optimization to fail. A solution may be to relax the target by achieving the minimum possible emissions. However this approach would force a deviation from the Net-Zero trajectory, and lead to high weight concentration undermining the portfolio's minimum-variance objective.

Covariance matrix estimation risk: The portfolio optimization relies on a covariance matrix constructed with 120 monthly observations ($\tau = 120$). However, the investment universe grows up to 579 firms ($N > \tau$). In this setting, the covariance matrix is highly unstable and susceptible to estimation error.

Transaction costs and survivorship bias: In the analysis it has been assumed a frictionless trading environment. However, since usually the Emerging Markets present lower liquidity and higher bid-ask spreads, actual returns would likely be lower. Furthermore, there may be a small degree of survivorship bias if some delisted firms are not correctly included in the dataset.

Return imputation bias (Zero-filling): Although assets with many missing data are excluded (at least 60 months data are required), some assets could still have up to 50% (60 months) missing observations. These remaining missing returns are replaced with exactly 0.0. This causes an artificially lower estimate of the volatility and covariances of these assets. The optimizations could then perceive these assets as safe and lead to over-allocations that are not justified by available data.

Uniform delisting assumption: In data cleaning, missing values at the end of a firm's dataset are considered as an indicator of default, imposing a return of -100% in the next month. This is a safe approach to manage bankruptcies, but it fails to consider the possibility of delisting due to mergers, acquisitions, or taking a company private.

7 Conclusion

This project studied whether financial performance of emerging market equities is not reduced when introducing carbon aware constraints into portfolios. The analysis used a dynamic investment universe, from 2014 to 2025, coupled with a rolling 10 year estimation windows and assigned Scope 1 and Scope 2 emissions data. With this setting, it compared a traditional portfolio allocations with more advanced and ambitious decarbonization strategies applied to new portfolios.

The initial comparison shows already a distinction between the two portfolios used as a benchmark. The value-weighted strategy achieved the highest performance, with a cumulative return of 131.34% and an annualized average return of 8.22%; but this performance comes with an higher volatility and downside risk. By comparison, the minimum-variance portfolio delivered lower returns but the project computed a smoother return path together with a lower volatility and a higher Sharpe Ratio. This confirm the variance minimization strategy as a defensive benchmark when for evaluating sustainability constraints.

In the second and main part of this project, carbon constraints' introduction did not materially lower portfolio performance. For an active investor choosing to impose a 50% carbon footprint reduction on its minimum-variance portfolio led to only losing 0.2%, from 6.72% to 6.52%. The same strategy weaken the volatility while the Sharpe Ratio remained practically unchanged. For the passive investor however, the tracking error portfolio with the 50% carbon reduction goal outperformed the benchmark, with an increased annualized return of 9.01% against the baseline of 8.22% and a light rise in volatility.

As shown in the results, a Net Zero strategy provides a strong evidence that a dynamic portfolio's decarbonization can be financially possible. By imposing a 10% annual reduction the carbon footprint, from 2013 the portfolio reduced its financed emissions by 77.5% by 2025. This strategy delivered an annualized return of 9.16% and a higher Sharpe Ratio than the value-weighted benchmark. Across all strategies, the main challenge was the reallocation of equities away from carbon driven sectors such as utilities, energy and materials towards more sustainable firms.

To conclude, the results provided by this report suggest that portfolio efficiency and the decarbonization can coexist in emerging markets equities. However, these results should be interpreted with caution as they may only represent a sector's dynamics in the time window considered. As future reference, a more comprehensive and complete analysis should include Scope 3 emissions, transaction costs, covariance shrinkage methods and even different market regimes in order to have a broader picture of possible financial benefits from low carbon allocation strategies.

8 AI Disclosure

For this report, a series of AI agents have been used (ChatGPT, Gemini, integrated Copilot in VSCode for Python) as support tools. This includes debugging parts of Python code (error messages, code readability) and controlling the english grammar of selected sections of this report. AI technologies were also used to discuss how to better present results and how to improve explanations. All methodological choices, portfolio decisions and results interpretation were made by the group in accordance to the rules provided.

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